

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2296

Canal Road 230 kV
July 01, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2296. The Interconnection Customer proposes a 358.2 MW battery energy storage facility interconnecting at the Canal Road 230 kV substation in the Ohio area. The request was received on March 13, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 615 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$97,642,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
W Millersburg-Canal Road 230 KV line	Weedpatch-Canal Rd 230 KV	125.8	574	48.7
Marys Rier-Knox 230 KV Line	El Paso-Ravenna 500 KV	124.1	573	5.2
W Salem-Newport Tie 100 KV	Weedpatch-Canal Road 230 KV	123.0	143	6.6
North Bay-Weedpatch 230 KV	Weedpatch-Portage Lakes 230 KV	117.0	418	11.5
Weedpatch 500 kV/230 kV Auto	Newport Tie-Post Rock 230 KV	110.6	783	29.4
Acme Hill-Seville 100 KV	Weedpatch-Canal Rd Substation 230 KV	110.3	183	5.7
Seville-Ryan 100 KV L	Weedpatch-Canal Road 230 KV	110.3	183	5.7
W Salem-Acme Hill 100 KV	Canal Rd-Weedpatch Substation 230 KV	109.3	192	5.7
N Bay-Sunnyside 230 KV	Portage Lakes-Weedpatch 230 KV line	109.0	412	11.5
Weedpatch-Canal Rd 230 KV Line	Beaver-Trego 500 KV	108.0	1,782	50.3
Knox-Atascocita 230 KV	Maple St-Atascocita 230 KV	105.4	487	5.5
Leetonia-J M Huber Substation 230 KV	El Paso-Ravenna 500 KV Line	104.8	751	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
West Millersburg-Canal Rd 230 KV	Rebuild line with higher-capacity conductor	\$45,445,000	42% (\$19,290,000)
Marys Rier-Knox 230 KV Line	Rebuild line with higher-capacity conductor	\$32,972,000	44% (\$14,461,000)
W Salem-Newport Tie 100 KV	Rebuild line with higher-capacity conductor	\$7,570,000	100% (\$7,570,000)
North Bay-Weedpatch 230 KV	Rebuild line with higher-capacity conductor	\$12,479,000	100% (\$12,479,000)
Weedpatch 500/230 kV Auto	Replace transformer with higher-rated unit	\$27,219,000	71% (\$19,436,000)
Seville-Acme Hill 100 KV	Rebuild line with higher-capacity conductor	\$4,093,000	70% (\$2,876,000)
Ryan-Seville 100 KV	Rebuild line with higher-capacity conductor	\$2,207,000	70% (\$1,551,000)
Acme Hill-W Salem Substation 100 KV	Reconductor line	\$4,153,000	70% (\$2,919,000)
N Bay-Sunnyside 230 KV	Reconductor line	\$9,904,000	100% (\$9,904,000)
Weedpatch-Canal Rd Substation 230 KV	Reconductor line	\$4,344,000	29% (\$1,276,000)
Knox-Atascocita 230 KV	Reconductor line	\$6,025,000	44% (\$2,668,000)
Leetonia-J M Huber 230 KV	Reconductor line	\$6,903,000	46% (\$3,212,000)
	Total allocated to project		\$97,642,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.